

Generalized Drinfeld-Sokolov Hierarchies II: The Hamiltonian Structures

Source-backed arXiv sample

Abstract

In this paper we examine the bi-Hamiltonian structure of the generalized KdV-hierarchies. We verify that both Hamiltonian structures take the form of Kirillov brackets on the Kac-Moody algebra, and that they define a coordinated system. Classical extended conformal algebras are obtained from the second Poisson bracket. In particular, we construct the W_n algebras, first discussed for the case $n=3$ and $l=2$ by A. Polyakov and M. Bershadsky.

1 Source Metadata

This sample is based on arXiv record `hep-th/9109014`. Authors: Nigel J. Burroughs, Mark F. deGroot, Timothy J. Hollowood and J. Luis Miramontes.

2 Extracted Prose 1

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3 Extracted Prose 2

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